ABSTRACT

Computer networks have a significant impact on the working of an organization. Universities depend on the proper functioning and analysis of their networks for education, administration, communication, e-library, automation, etc. An efficient network is essential to facilitate the systematic and cost-efficient transfer of information in an organization in the form of messages, files, and resources. The project provides insights into various concepts such as topology design, IP address configuration, and how to send information in the form of packets to the wireless networks of different areas of a university.

The aim of this project is to design the topology of the university network using the software Cisco Packet Tracer with the implementation of wireless networking systems. This universitynetwork consists of the following devices:

1) Router (1941)
2) Switches (2960-24TT)
3) WEB server (HTTP)
4) PCs
5) Laptops

The design includes the following parts of the University:

Main Campus:

6) **Level 1:** Houses the departments of Management, HR, and Finance, with administrative staff PCs distributed throughout offices. Use VLANs to segment departmental traffic as they share networking equipment. This floor also includes a Computer Lab for students.

7) **Level 2:** Library

8) **Level 3:** Contains Student Labs, the Lecturer’s Room, and the IT Department, which hosts the university's web server and other critical servers.

Smaller Campus:

9) **Faculty of Health and Science:** Staff offices and student labs are located on separate floors.

Table of Contents

INTRODUCTION 8

Motivation 8

Project Statement 9

LITERATURE REVIEW 9

What is Packet Tracer? 9

- Router 10
- Switch 10
- Network Packet 10
- Server 10

➢ DNS Server 11

➢ WEB Server 11

- Ethernet 11
- Computing Device 11
- Internet Protocol 12
- Simulation Environment 12

WORK DONE 13

1. Software and hardware requirements 13
2. Network Requirements 13
3. Implementation and Flow Diagram 18
1. Configuring IP Addresses 20

- Level 1 23

Management 23

- HR 24
- Computer Lab 25
- Library 25

Level 3 27

- Student's Labs 27
- Lecturer's Room 27

Conclusion 36

- Future Work 36

INTRODUCTION

Motivation

The word “digital” is very significant in today’s world, with an increase in the developmentof technology the entire world is moving towards the digital era. The educational institution plays an important role in this digitalization; hence the campus should adapt to digital means of networking as well and become a “digital campus”. Going wireless plays an important rolein this digitalization. The wireless network makes the connection easy with a reduction in the use of wires or cables. A wired connection makes it difficult to keep track of all the devices and to manage the cable connection, which is not only chaotic but also challenging to handle.

Campus networking via wireless connection becomes an important part of campus life and provides the main way for teachers and students to access educational resources, which gives an important platform to exchange information. As laptops and intelligent terminals arewidely used, demand for access to information anytime and anywhere has become more and more urgent, but traditional cable networks cannot meet this requirement. Then wireless network construction becomes necessary and essential. The wireless network is one of the important components of a digital campus and wisdom campus. It provides an efficient wayto explore the internet with a mobile terminal for teachers and students regardless of cables and places.

This is an important mark of the modern campus as a supplement of a cable network. Withthe development of network and communication technology, cable networks on a university campus bring much convenience for teaching and research work. But for mobility andflexibility, it has obvious shortcomings. A wireless network can overcome these drawbacks and has been applied to the university campus.

Project Statement

In this project, we defined a simulation of campus networks based on network topology. The network is divided into two sets: one for the main campus area and the other for the smaller campus.

The major aim of this project is to show the network connectivity that is used in universitiesto make the network efficient and mobile at the same time. Mobility is the major concentration of this project. In order to provide equal functionality to all the users (college staff and students), we have added DNS, HTTP servers for the maximum utilization of resources.

Hence the campus network provides different services such as connecting the user to the internet, data sharing among users (students, teachers, and different university members), accessing different web services for different functionalities, so it needs wireless networking for smooth processing.

LITERATURE REVIEW

What is Packet Tracer?

Packet Tracer is a cross-platform visual simulation tool designed by Cisco Systems thatallows users to create network topologies and imitate modern computer networks. The software allows users to simulate the configuration of Cisco routers and switches using a simulated command-line interface. Packet Tracer makes use of a drag-and-drop user interface, allowing users to add and

remove simulated network devices as they see fit. Thesoftware is mainly focused on Certified Cisco Network Associate Academy students as an educational tool for helping them learn fundamental CCNA concepts. Previously students enrolled in a CCNA Academy program could freely download and use the tool free of charge for educational use.

- **Router**

  A router is a device like a switch that routes data packets based on their IP addresses. The router is mainly a Network Layer device. Routers normally connect LANs and WANstogether and have a dynamically updating routing table based on which they make decisions on routing the data packets. Router divides broadcast domains of hosts connected through it.

- **Switch**

  A network switch (also called switching hub, bridging hub, officially MAC bridge is networking hardware that connects devices on a computer network by using packet switchingto receive and forward data to the destination device. A network switch is a multiportnetwork bridge that uses MAC addresses to forward data at the data link layer (layer 2) of theOSI model. Some switches can also forward data at the network layer (layer 3) by additionally incorporating routing functionality. Such switches are commonly known aslayer-3 switches or multilayer switches.

- **Network Packet**

  A network packet is a formatted unit of data carried by a packet-switched network. A packet consists of control information and user data, which is also known as the payload.

- **Server**

  A server is a computer or system that provides resources, data, services, or programs to other computers, known as clients, over a network. In theory, whenever computers share resources with client machines, they are considered servers. There are many types of servers, including web

servers, mail servers, and virtual servers.

Many networks contain one or more of the common servers. The servers used in our project are as follows:

➢ **DNS Server**

DNS stands for Domain Name System servers which are application servers that provide a human-friendly naming method to the user computers to make IP addresses readable by users. The DNS system is a widely distributed database of names and other DNS servers, each of which can be used to request an otherwise unknown computer name. When a user needs the address of a system, it sends a DNS request with the name of the desired resource to a DNS server. The DNS server responds with the necessary IP address from its table of names.

➢ **WEB Server**

One of the widely used servers in today's market is a web server. A web server is a special kind of application server that hosts programs and data requested by users across the Internet or an intranet. Web servers respond to requests from browsers running on client computers for web pages, or other web-based services.

- **Ethernet**

This is the backbone of our network. It consists of the cabling and is typically able to Transfer data at a rate of 100mb/s. It is a system for connecting a number of computersystems to form a local area network, with protocols to control the passing of information andto avoid simultaneous transmission by two or more systems. Among the different types ofethernet, we have used Gigabit Ethernet, which is a type of Ethernet network capable oftransferring data at a rate of 1000 Mbps and fast Ethernet is a type of Ethernet network thatcan transfer data at a rate of 100 Mbps.

- **Computing Device**

Computing devices are the electronic devices that take user inputs, process the inputs, and then provide us with the end results. These devices may be Smartphones, PC Desktops, Laptops, printer, and many more.

- **Internet Protocol**

  Internet Protocol (IP) is one of the fundamental protocols that allow the internet to work. IP addresses are a unique set of numbers on each network, and they allow machines to address each other across a network. It is implemented on the internet layer in the IP/TCP model.

- **Simulation Environment**

  The simulations of our network topology can be easily achieved using cisco packet tracer. Using a simulation mode, you can see packets flowing from one node to another and can also click on a packet to see detailed information about the OSI layers of the networking. Packet Tracer offers a huge platform to combine realistic simulation and visualize themsimultaneously. Cisco Packet Tracer makes learning and teaching significantly easier by supporting multi-user collaboration and by providing a realistic simulation environment for experimenting with projects.

WORK DONE

To make our project understandable, we have divided the content into steps. They are as follows:

1. Software and hardware requirements

Before heading towards the implementation, we need to make sure of the following requirements.

- A proper workstation (any mid-high range laptop will suffice).
- Packet Tracer by Cisco
- 8 GB RAM.
- Any 10,000+ Average CPU Mark scored processor.
- 256 GB of SSD
- USB 3.0+ port.

2. Network Requirements

City University Malaysia outline is considered for this wireless university network.The network is divided into 2 areas.
For a comprehensive report and presentation on your network design, here are some additional data and details you can include, based on your setup:

1. Network Topology Diagram
Show the hierarchical structure with core switches, routers, and end devices (PCs, printers, laptops).
Highlight the VLAN segments (VLAN 10 to VLAN 100).
Indicate the interconnections between switches and the router (e.g., trunking, routing interfaces).
Demonstrate the use of a multilayer switch for routing between VLANs.
2. VLAN Configuration

Purpose: VLANs are used to segment the network to improve security and performance.

VLAN 10: Management - IP Range: 192.168.10.0/24, Gateway: 192.168.10.1
VLAN 20: HR - IP Range: 192.168.20.0/24, Gateway: 192.168.20.1
VLAN 30: Finance - IP Range: 192.168.30.0/24, Gateway: 192.168.30.1
VLAN 40: Computer Labs - IP Range: 192.168.40.0/24, Gateway: 192.168.40.1
VLAN 50: Library - IP Range: 192.168.50.0/24, Gateway: 192.168.50.1
VLAN 60: Student Labs - IP Range: 192.168.60.0/24, Gateway: 192.168.60.1
VLAN 70: Lecturers' Room - IP Range: 192.168.70.0/24, Gateway: 192.168.70.1
VLAN 80: IT Department - IP Range: 192.168.80.0/24, Gateway: 192.168.80.1
VLAN 90: Student's Lab - IP Range: 192.168.90.0/24, Gateway: 192.168.90.1
VLAN 100: Staff Offices - IP Range: 192.168.100.0/24, Gateway: 192.168.100.1
3. Routing Configuration (RIP and Inter-VLAN Routing)
RIP Configuration:
RIP v2 is used for routing between the different VLANs in your network.
Example configuration on the router:
bash
Copy code
router rip
version 2
network 192.168.0.0
no auto-summary
Subinterfaces: Each VLAN is routed through subinterfaces on the router, such as:
bash
Copy code
interface gig0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0

interface gig0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
4. DHCP Configuration
The router provides DHCP services to each VLAN. Each VLAN has its own DHCP pool.

bash

Copy code

ip dhcp pool Management

network 192.168.10.0 255.255.255.0

default-router 192.168.10.1

dns-server 192.168.10.1

exit

Example for each VLAN (HR, Finance, etc.).

5. Security Configuration

VLAN Access Control: Use access control lists (ACLs) to restrict access between VLANs where necessary.

Port Security: Configure port security on switches to restrict the number of MAC addresses allowed per port.

Example:

bash

Copy code

switchport port-security maximum 2

switchport port-security violation restrict

switchport port-security

6. Server Configuration

Web Server: Host internal resources like intranet, department pages, and other services.

IP Address: 192.168.80.10, Gateway: 192.168.80.1

FTP Server: Provide file-sharing services for users in different VLANs.

IP Address: 192.168.80.11, Gateway: 192.168.80.1

7. Network Testing

Ping Tests: Test connectivity between devices within the same VLAN and across VLANs (Inter-VLAN routing).

Example:

bash

Copy code

ping 192.168.10.10 # Ping a device in VLAN 10

Traceroute: Use traceroute to ensure proper routing between VLANs.

Example:

bash

Copy code

```
traceroute 192.168.20.10  # Trace route to a device in VLAN 20
```

Simulation Mode in Cisco Packet Tracer: Use Cisco Packet Tracer's simulation mode to visualize and diagnose network traffic, ensuring that configurations are working properly.

8. Future Scalability

The design supports easy expansion by adding more VLANs for future departments or faculties.

Using hierarchical and modular design principles (Core, Distribution, and Access layers) allows for easier management and scalability.

9. Security Measures

VLAN Segmentation: Helps in isolating traffic between departments, improving security and reducing congestion.

DHCP Snooping: To prevent unauthorized DHCP servers from assigning IP addresses to devices.

Port Security: Restricting which devices can connect to which ports to prevent unauthorized access.

10. Report and Presentation Outline

Introduction: Brief overview of the network design, its purpose, and the key components.

Network Topology: Display and explain the network diagram, VLANs, and IP addressing scheme.

VLAN and Routing: Discuss the configuration of VLANs and inter-VLAN routing.

DHCP Configuration: Explain how DHCP is used to allocate IP addresses dynamically.

Security Configuration: Describe the security measures implemented on switches and routers.

Testing and Troubleshooting: Provide results from tests like ping, traceroute, and simulation mode.

Future Scalability and Conclusion: Discuss how the network can expand in the future and summarize the design.

Presentation Slides Tips:

Slide 1: Title Slide (Network Design for City University)

Slide 2: Network Overview (Brief explanation of the requirements)

Slide 3: Network Topology Diagram (VLANs, routers, and switches)

Slide 4: VLAN Configuration (List VLANs, IP ranges, and purpose)

Slide 5: Routing and DHCP Configuration (RIP routing, subinterfaces, DHCP pools)

Slide 6: Security and Testing (Port security, ACLs, and test results)

Slide 7: Scalability and Future Expansion (How the design can grow)

Slide 8: Conclusion and Final Thoughts

By using this data and structure, you can develop a detailed and thorough report as well as a professional presentation.

1. Main Campus Area

Level 1: Houses the departments of Management, HR, and Finance, with administrative staff PCs distributed throughout offices. Use VLANs to segment departmental traffic as they share networking equipment. This floor also includes a Computer Lab for students.

Level 2: Library

Level 3: Contains Student Labs, the Lecturer’s Room, and the IT Department, which hosts the university's web server and other critical servers.

2. Smaller Campus Area

Faculty of Health and Science: Staff offices and student labs are located on separate floors.

Devices Used in The Network:

Devices	Quantity
1) Router (2911)	2
2) Switches (2960-24TT)	10
3) Multilayer Switch	2
5) WEB server (HTTP)	1
6) FTP server	7
7) PCs	33
8) Laptops	4
9) Printers	13

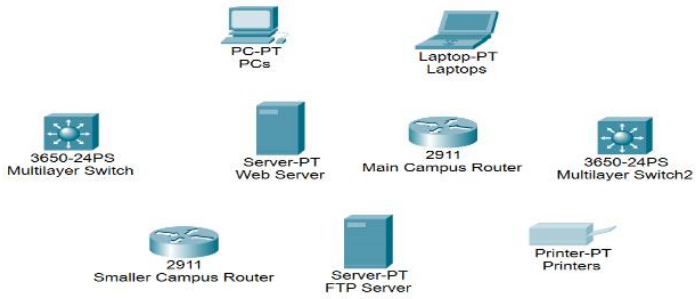
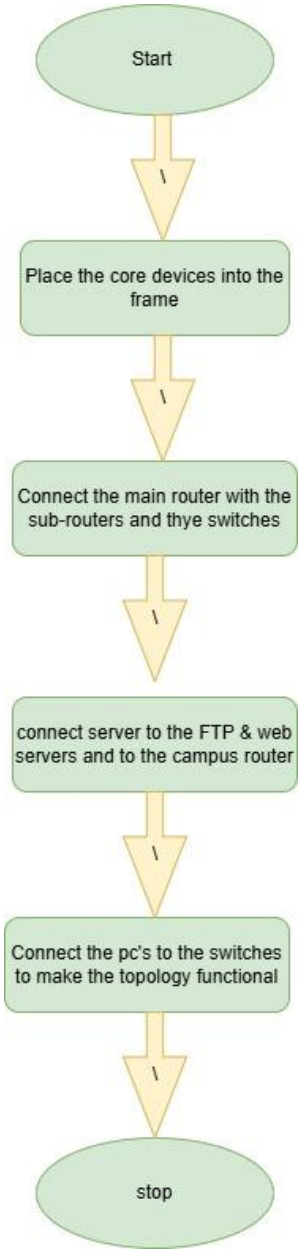


Figure 3: Devices used in the network.

3. Implementation and Flow Diagram

- To design the wireless network of the university we initially started by placingthe core devices into the frame as mentioned in the layout.
- Firstly, we placed the **main router** at the center of the university outline, which was further connected to the **server switch** using the gigabit ethernet port with copper straight-through cable and sub routers (**campus router and hostel router**) using the serial port with serial DCE cable at the hostel areaand campus area respectively.
- The server switch was further connected to the **FTP and WEB**

  servers respectively.
- The wireless access points were then connected to computing devices (PCs, laptops, and Printers).
- All these connections are made through ethernet ports (gigabit ethernet andfast ethernet) using copper straight-through cables.

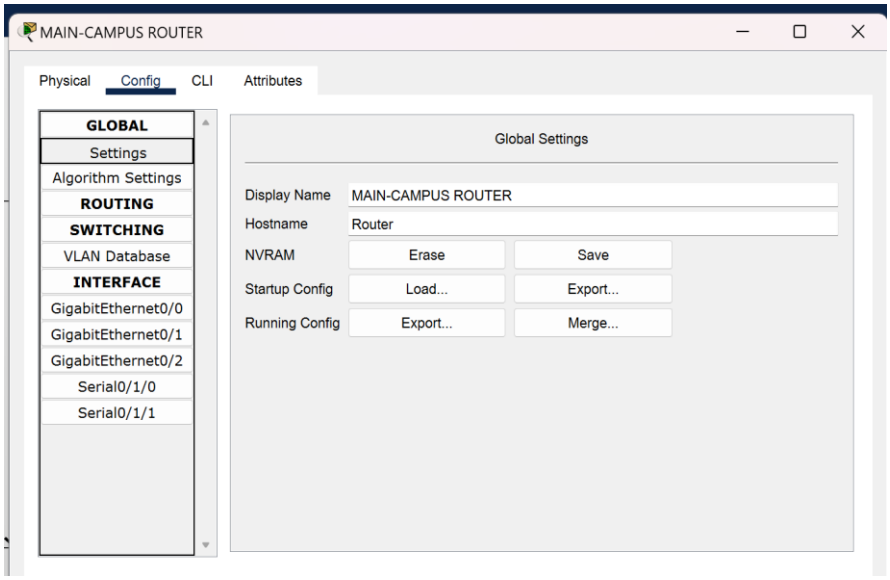


This is the flow diagram for a better understanding of the steps mentioned above.

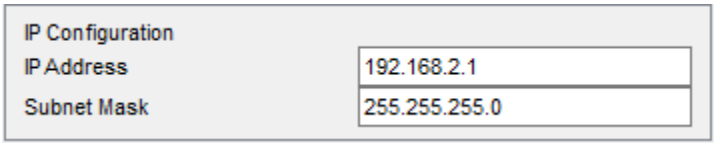
1. Configuring IP Addresses

We have attached the screenshots of all the IP configuration below:

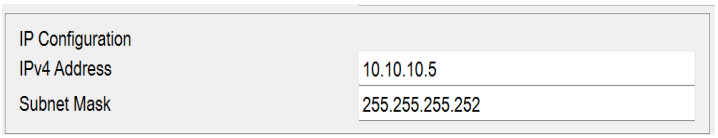
- Main Router configuration



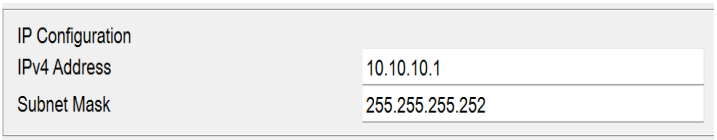
GigabitEthernet0/1



Serial0/1/0



Serial0/1/1

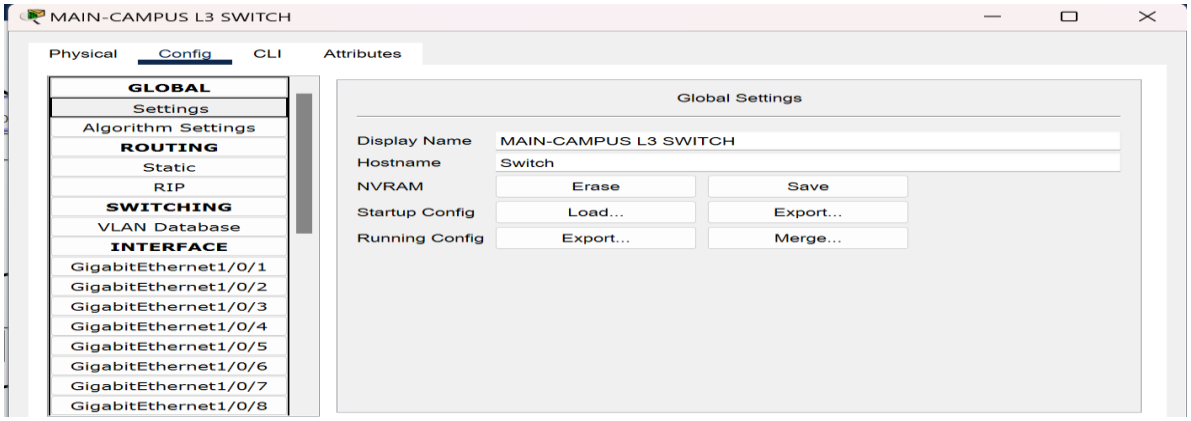


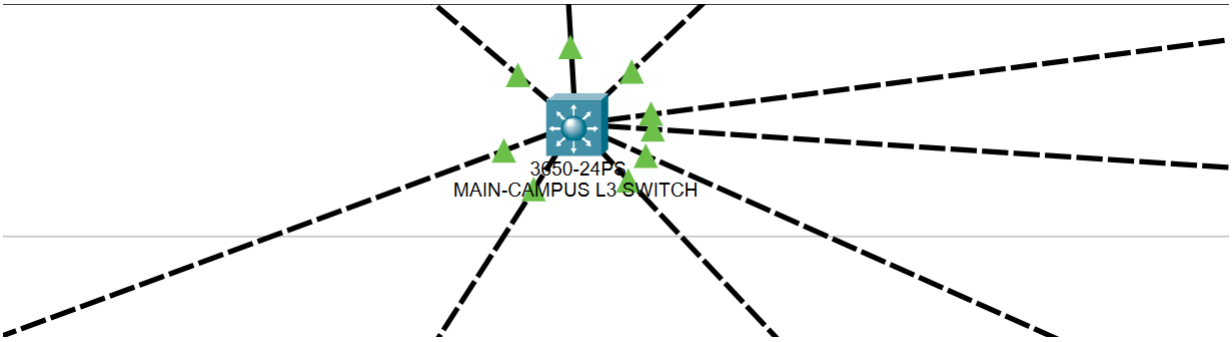
RIP

Network Address
10.0.0.0
192.168.1.0
192.168.2.0
192.168.3.0
192.168.4.0
192.168.5.0
192.168.6.0
192.168.7.0
192.168.8.0
192.168.10.0
192.168.20.0
192.168.30.0

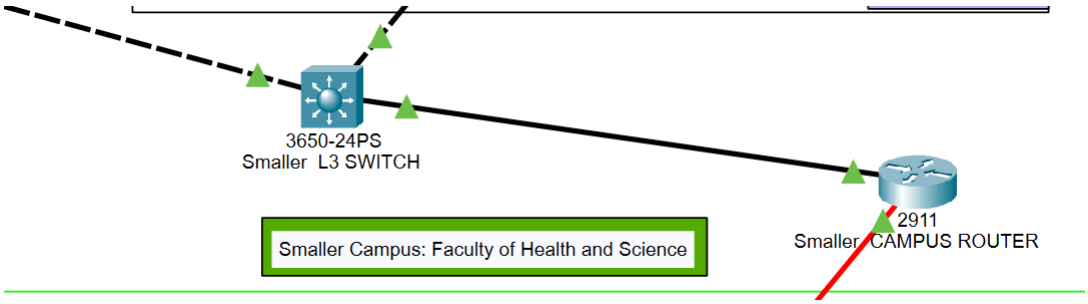
192.168.40.0
192.168.50.0
192.168.60.0
192.168.70.0
192.168.80.0

Multilayer Switch Main Campus

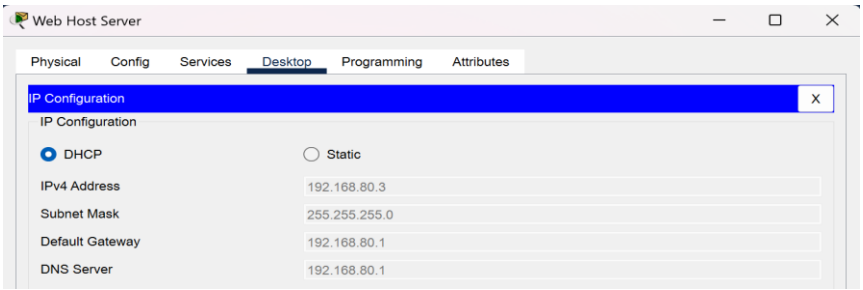


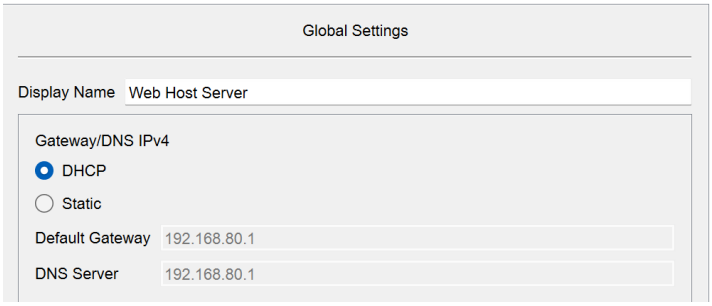


Multilayer Switch Smaller Campus

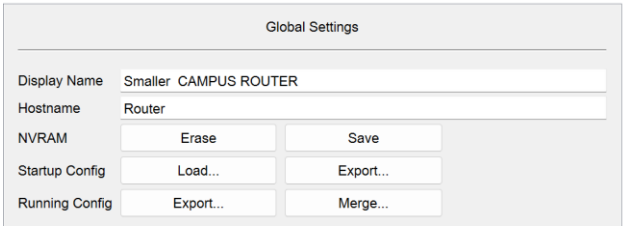


- **WEB SERVER**

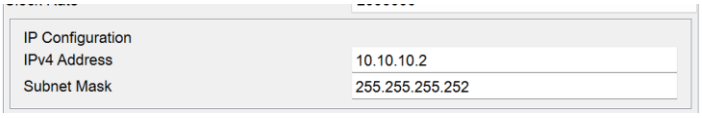




- **SMALLER CAMPUS ROUTER**



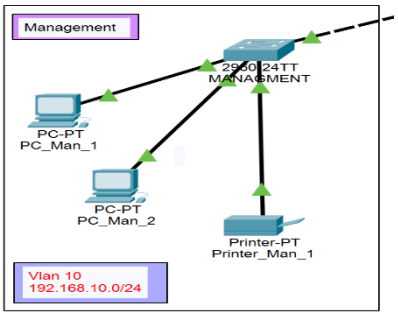
Serial0/2/0



Main Campus

- **Level 1**

Management



IP Address are as follows192.168.10.2 - PC

192.168.10.4 - PC

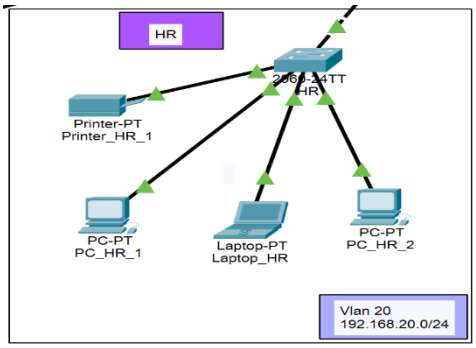
192.168.10.1 - Printer

Subnet Mask- 255.255.255.0

Default Gateway- 192.168.10.1

DNS Server- 192.168.10.1

- **HR**



IP Address are as follows192.168.20.2 - PC

192.168.20.3 - Laptop

192.168.20.4 - PC

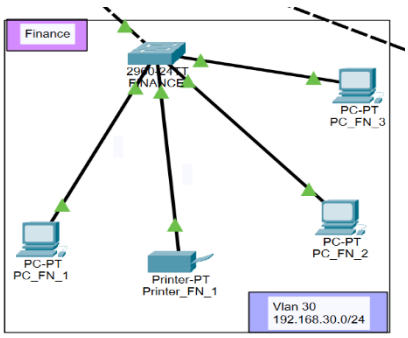
192.168.20.1- Printer

Subnet Mask- 255.255.255.0

Default Gateway- 192.168.20.1

DNS Server- 192.168.20.1

Finance



IP Addresses are as follows192.168.30.5- PC

192.168.30.4- Printer

192.168.30.3- PC

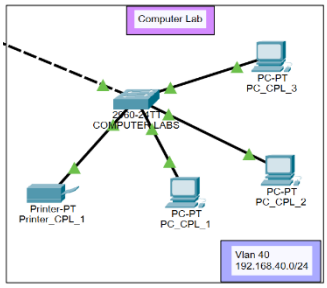
192.168.30.2- PC

Subnet Mask- 255.255.255.0

Default Gateway- 192.168.30.1

DNS Server- 192.168.30.1

- **Computer Lab**



IP Addresses are as follows192.168.40.2- PC

192.168.40.3- PC

192.168.40.4- PC

192.168.40.6- Printer

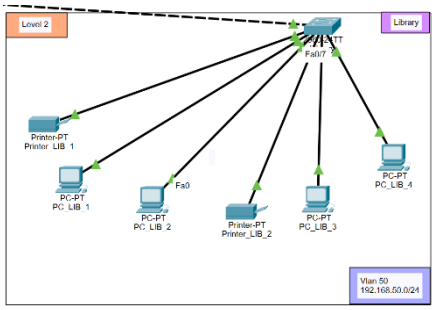
Subnet Mask- 255.255.255.0

Default Gateway- 192.168.40.1

DNS Server- 192.168.40.1

Level 02

- **Library**



IP Addresses are as follows192.168.50.2- PC

192.168.50.3- PC

192.168.50.4- Printer

192.168.50.5- Printer

192.168.50.6- PC

192.168.50.7- PC

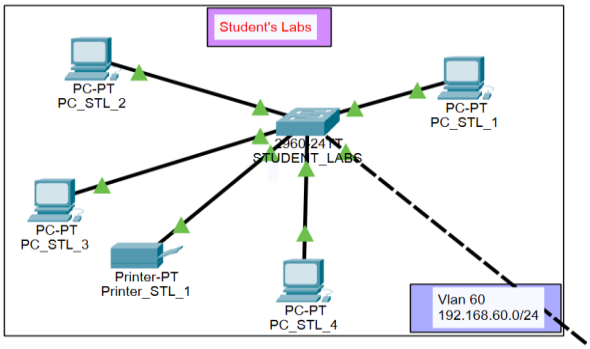
Subnet Mask- 255.255.255.0

Default Gateway- 192.168.50.1

DNS Server- 192.168.50.1

Level 3

- **Student's Labs**



IP Addresses are as follows192.168.60.1- Printer

192.168.60.2- PC

192.168.60.3- PC

192.168.60.4- PC

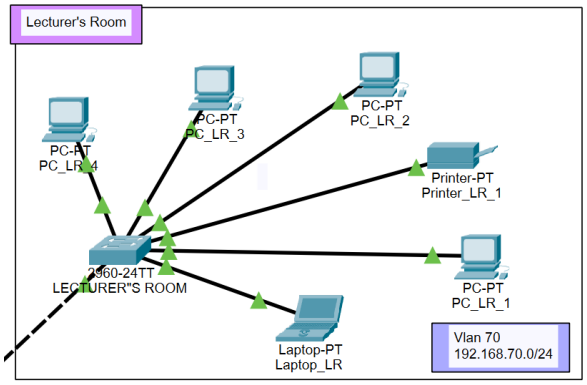
192.168.60.5- PC

Subnet Mask- 255.255.255.0

Default Gateway- 192.168.60.1

DNS Server- 192.168.60.1

- **Lecturer's Room**



IP Addresses are as

follows192.168.70.2- PC

192.168.70.3- PC

192.168.70.4- PC

192.168.70.5- Laptop

192.168.70.6- PC

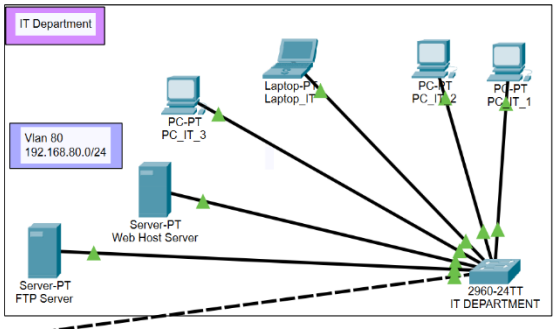
192.168.70.7 - Printer

Subnet Mask- 255.255.255.0

Default Gateway- 192.168.70.1

DNS Server- 192.168.70.1

It Department



IP Addresses are as

follows192.168.80.2- PC

192.168.80.3- PC

192.168.80.4- PC

192.168.80.5- Laptop

192.168.80.6- PC

192.168.80.7 - Printer

Subnet Mask- 255.255.255.0

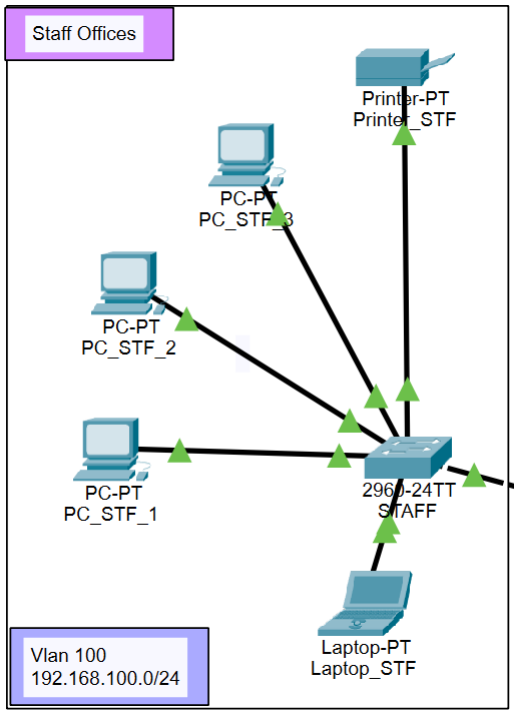
Default Gateway- 192.168.80.1

DNS Server- 192.168.80.1

Smaller CAMPUS

Floor 1

Staff Offices



IP Addresses are as follows192.168.100.3- PC

192.168.100.4- PC

192.168.100.5- PC

192.168.100.6- Printer

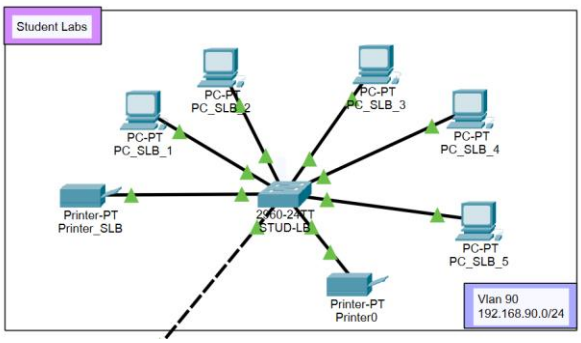
Subnet Mask- 255.255.255.0

Default Gateway- 192.168.100.1

DNS Server- 192.168.100.1

Floor 02

Student Labs



IP Addresses are as follows192.168.90.2- PC

192.168.90.3- PC

192.168.90.4- PC

192.168.90.6- Laptop

192.168.90.7- PC

192.168.90.8 – Printer

192.168.90.5 - Printer

Subnet Mask- 255.255.255.0

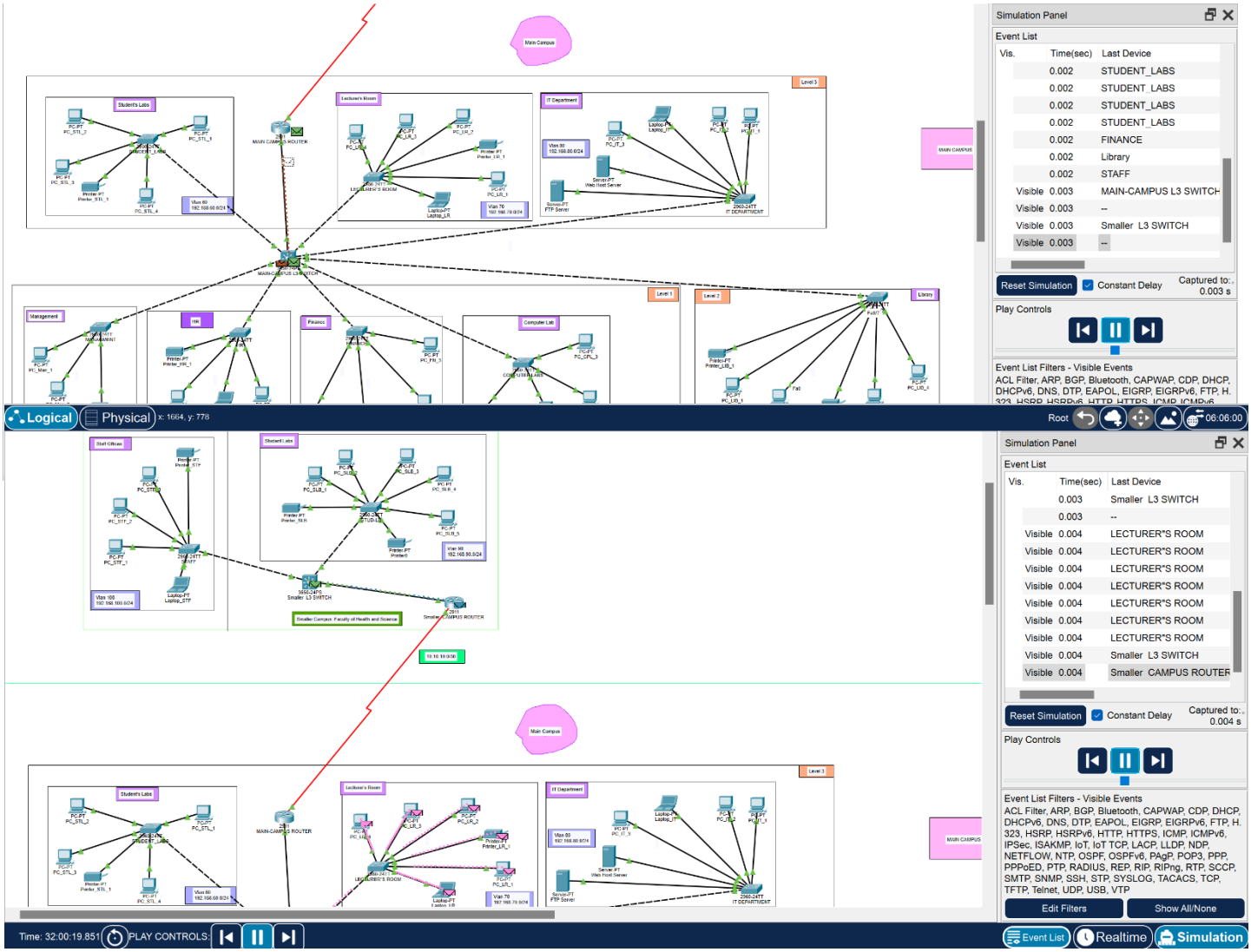
Default Gateway- 192.168.90.1

DNS Server- 192.168.90.1

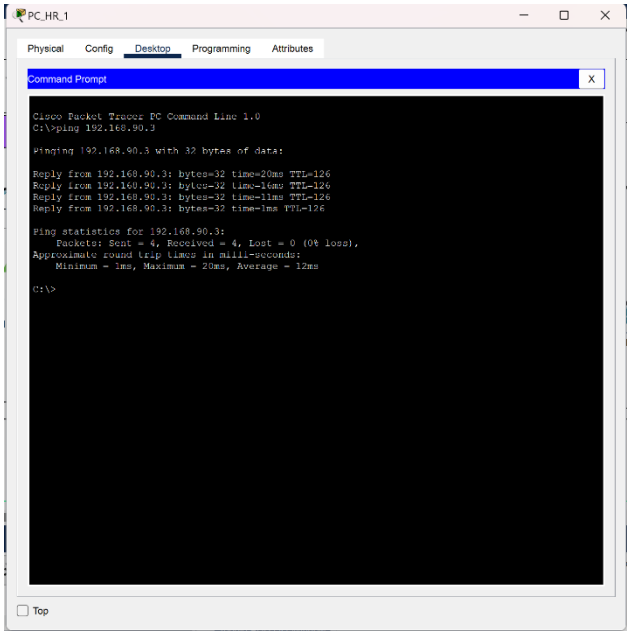
- **Final Simulation**

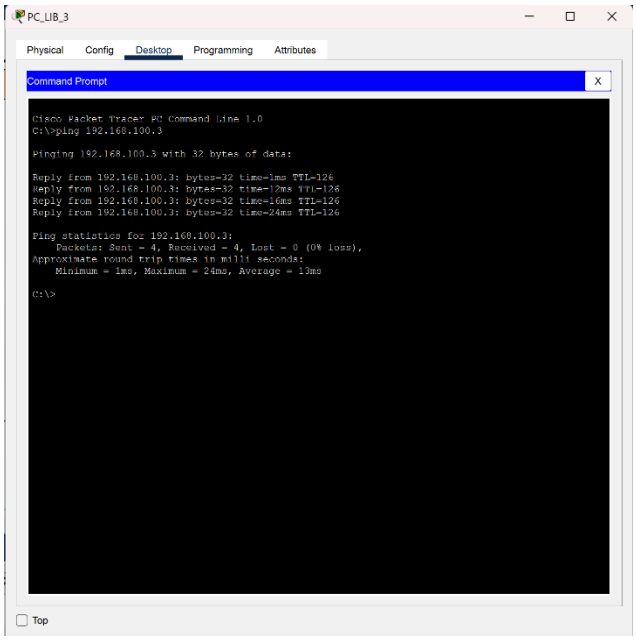
In Simulation Mode, you can watch your network run at a slower pace, observing the paths that packets take and inspecting them in detail. The proposed architecture, when simulated on Cisco Packet Tracer, produced results which are demonstrated as follows:

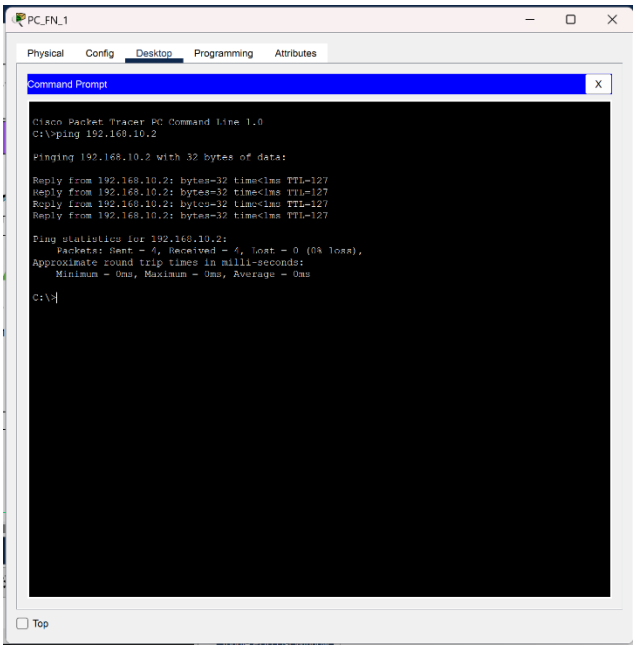
Final simulation for the network system to check all the connections

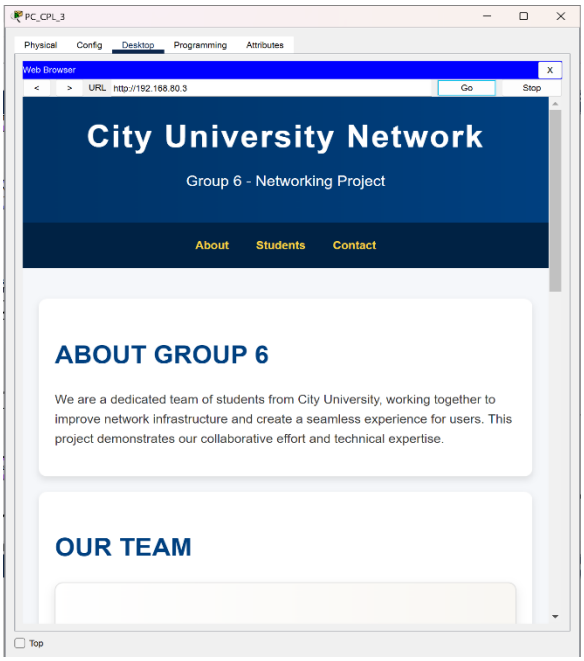


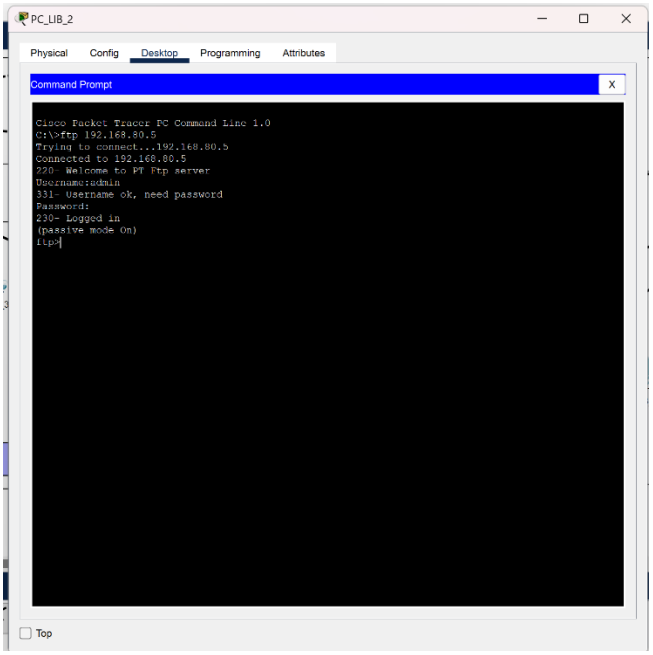
- **Ping Test:** Network connectivity and communication can be tested using the ping command, followed by the domain name or the IP address of the device (equipment) whose connectivity one wishes to verify.





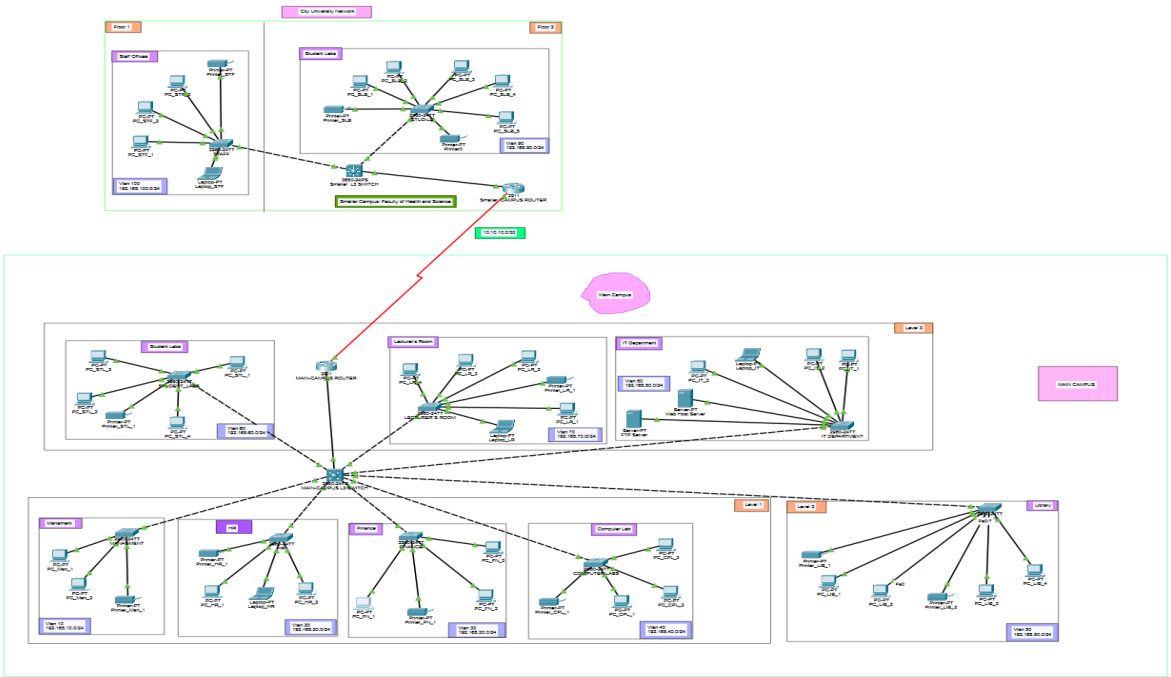






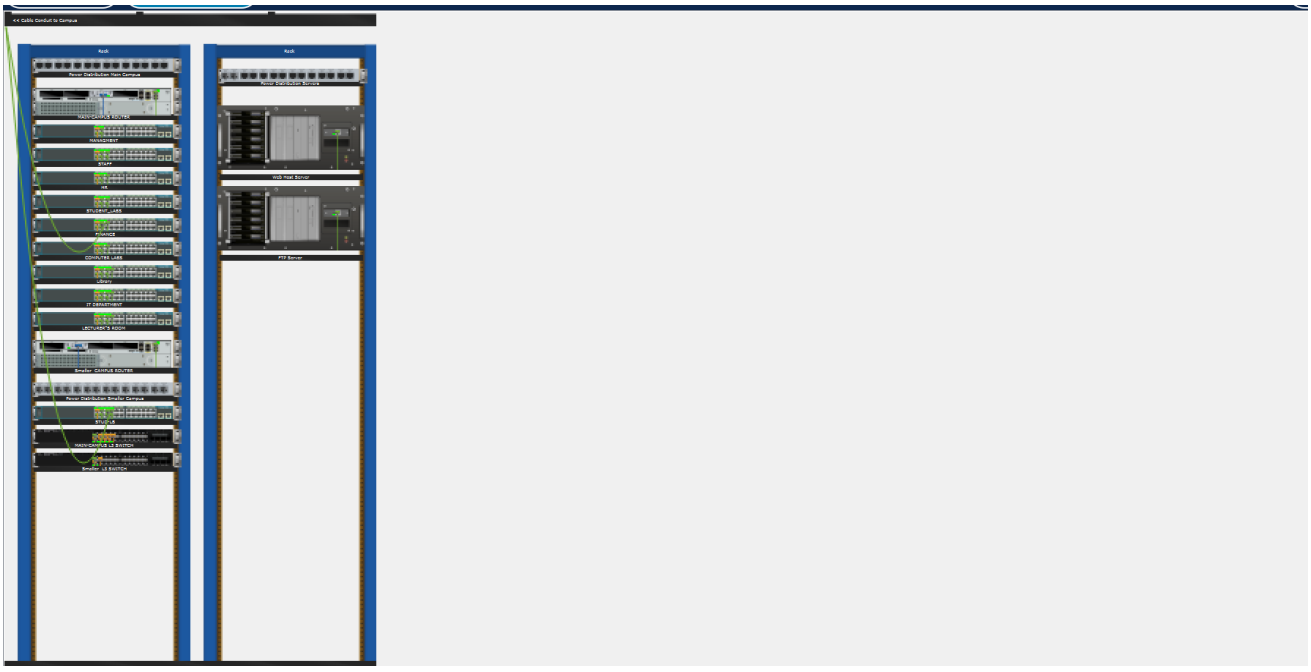
SCREENSHOTS –

Logical View

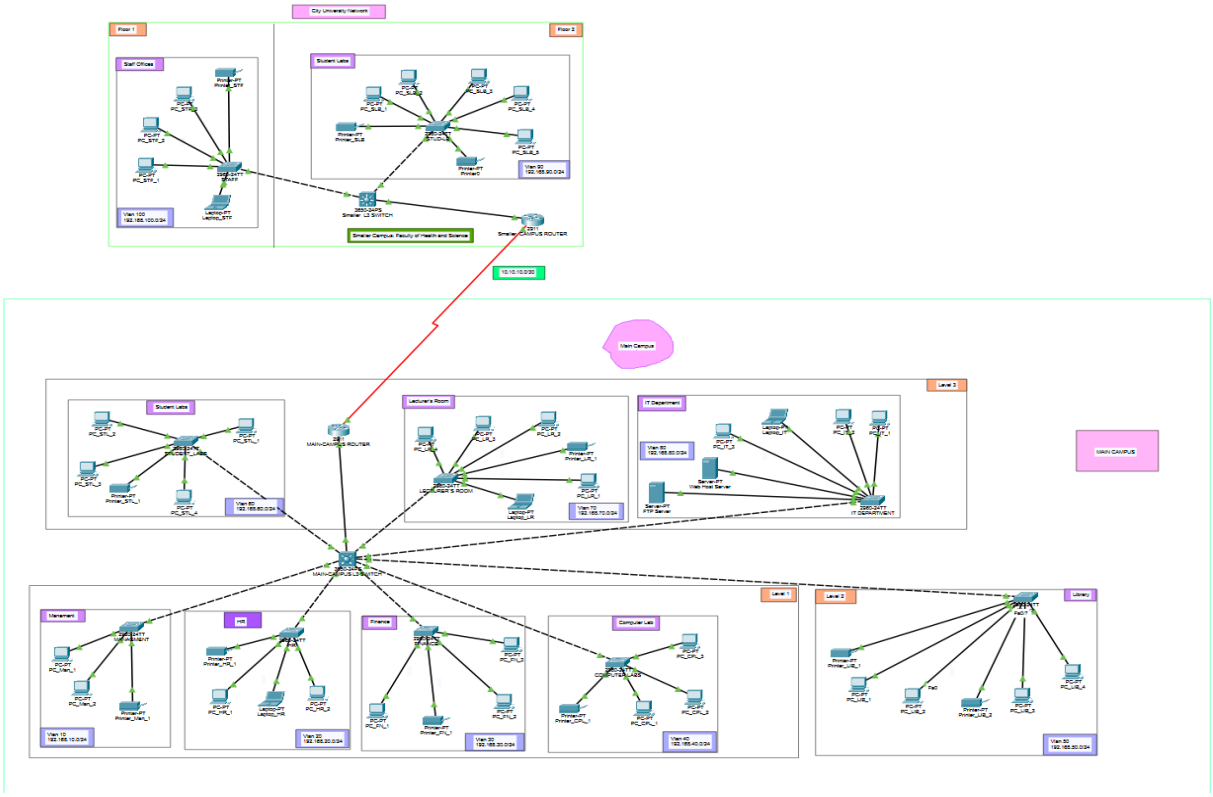


Racks –


Rack
Power Distribution Main Campus
MAIN-CAMPUS ROUTER
MANAGMENT
STAFF
HR
STUDENT LABS
Rack
Power Distribution Servers
Web Host Server



Final Network Design –



Conclusion

We started our discussion with the word “digitalization” and in order to achieve it, we aimed to start with an educational institute, and finally, we designed a network for a university. As we mentioned, mobility and efficiency are the key aspects of wireless networks, which were our main goal, and hence, we decided toshift to a wireless network instead of a wired one, making our network clean and less chaotic.

In this project, we designed a University Network using Cisco Packet Tracerthat uses a networking topology implemented using servers, routers, switches, and enddevices in a multiple area network. We have covered all the necessary features that are required for a network to function properly. We have included a DNS server and a web server for establishing a smooth communication system between different areasof our network and specifically for the communication between students and teachers.

- **Future Work**

The configuration and specifications are for the initial prototype and can further be developedand additional functionality can be added to increase support and coverage of our existing network.

REFERENCES

[1] https://en.wikipedia.org/wiki/Packet_Tracer

[2] https://www.paessler.com/it-explained/server

[3] https://computernetworking747640215.wordpress.com/2018/07/05/secure-shell-ssh-configuration-on-a-switch-and-router-in-packet-tracer/

[4] http://router.over-blog.com/article-how-to-configure-cisco-router-password-106850439.html

[5] https://www.cognoscape.com/benefits-going-wireless/